Hydrophone digital board

Feedback for Jolie

Note from Eli

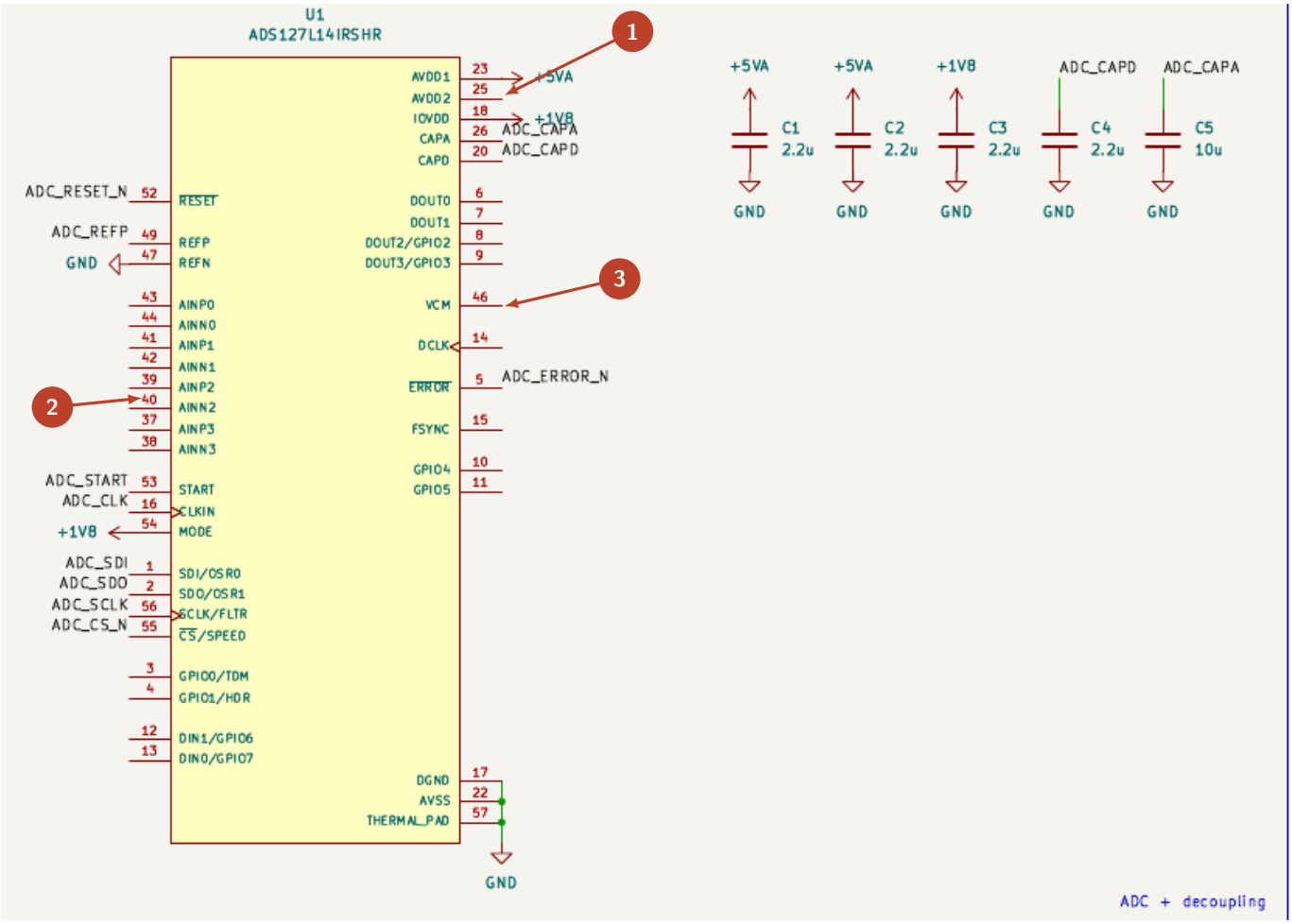
"This is an audit I had my bot performed. I looked over it's results and checked for any mistakes it made and iterated a few times on the feedback. Not saying this is perfect but I think it should catch any of the obvious critical issues, some of which I confirmed manually"

ID	Feedback item	Classification
D1	Supply U1 AVDD2 pin 25 and provide local bypassing	Required completion
D2	Complete differential inputs and common-mode interface	Integration gap
D3	Connect and demonstrate the ADC sample-data receiver	Integration gap
D4	Complete connector/power interfaces and repair hierarchy	Integration gap
D5	Verify reference stability, pin defaults and recording rate	Qualification checks

Scope: Receive-only, work in progress. Open connections are items to finish. The sample receiver and common-mode source remain team decisions; D5 lists checks, not proven failures.

D1–D2 / ADC supply and analog interface

Confirmed connections still to complete



1 / D1 — AVDD2 is isolated. Netlist entry `unconnected-(U1-AVDD2-Pad25)` contains U1.25 only. AVDD1 pins 23/24 are powered; that does not supply the separate modulator rail. Select a permitted AVDD2 rail and add local bypassing. TI permits tying AVDD2 to AVDD1; its bypass table specifies 2.2 μF for AVDD2.

2 / D2 — All four input pairs are open. U1 pins 37–44 are individual unconnected nets. Complete the CH0–CH3 differential connections through a matched board connector and the chosen input RC/protection network. Verify polarity, common-mode range and overload behavior jointly with the analog board.

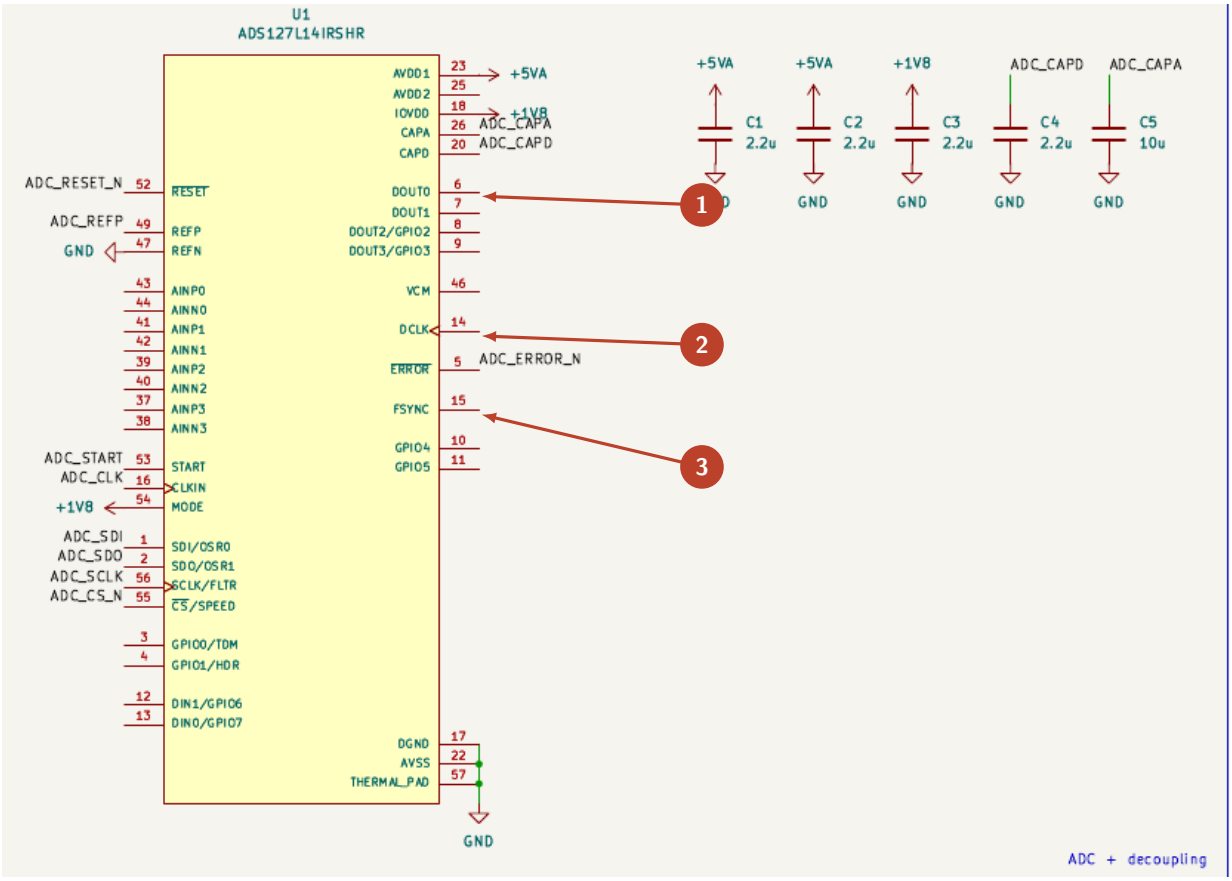
3 / D2 — VCM is unconnected. Pin 46 does not reach the analog board's VOCM interface. Team decision: choose the DC center voltage around which each differential signal swings, and which circuit supplies it. The ADC VCM output is one candidate. VCM may legally be unused if disabled; an open pin alone is not a fault.

Acceptance. Updated netlist shows U1.25 on its supply and bypass network, eight mapped analog inputs, and an explicit common-mode strategy. Check effective capacitor values and physical placement during BOM/layout review.

Source: TI ADS127L14/18: pin functions, VCM and supply/bypass requirements.

D3 / Complete the actual sample path

Confirmed • Register SPI is not sample capture



1–3 — Sample receiver remains to be designed. DOUT0–3 (pins 6–9), DCLK (14) and FSYNC (15) are all isolated. The wired SDI/SDO/SCLK/CS interface configures registers; ADC samples use the separate frame-sync port. The team has not demonstrated capture yet. Choose lane count, framing, receiver pins and 1.8-to-3.3 V translation as one design.

Check unused pins by mode. DIN0/1 (13/12) and other multiplexed GPIO pins are open. After choosing the data-port mode, define their reset and operating states. TI requires unused DIN inputs not to float; not every unused DOUT/GPIO must be wired as a data lane.

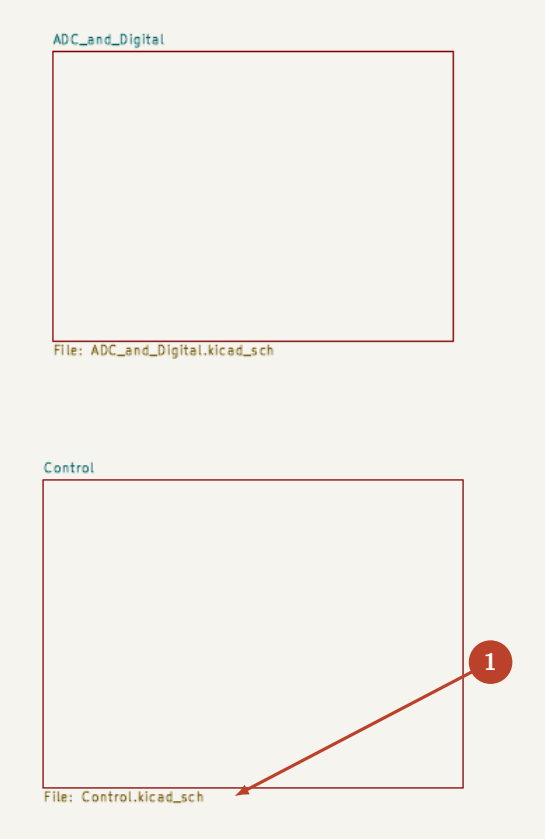
Clock is not a proven fault. The 65.536 MHz oscillator can support a divide-by-two ADC-core configuration. Prove the exact clock divider, data-clock timing, lane format and host DMA setup together; do not replace the oscillator solely because the core target is 32.768 MHz.

Acceptance. Demonstrate four synchronous channels at the spec's 512 ksample/s/channel target using the selected receiver or bridge. A successful register read is not evidence of sample capture. The current files do not establish that Daisy can sustain this configuration.

Source: TI ADS127L14/18: frame-sync data port, DIN defaults and clock configuration.

D4 / Finish the board boundary

Confirmed • Shared integration responsibility



1 — Stale sheet reference. The digital root still names `Control.kicad_sch`, but that file is absent from this project after the board split. Remove the stale sheet or restore the intended architecture. Regenerate the hierarchy, netlist and schematic PDF without missing-sheet warnings.

No paired connector yet. The digital project contains the microSD connector and Daisy module, but no mate to analog J5. Local I2C/reset labels on Daisy do not constitute an inter-board connection.

Supply interface unfinished. +5V at Daisy VIN and +5VA at the ADC/reference are separate nets without an implemented input connector/supply source in this project. Define how each is powered and budget the downstream +3.3V and +1V8 loads.

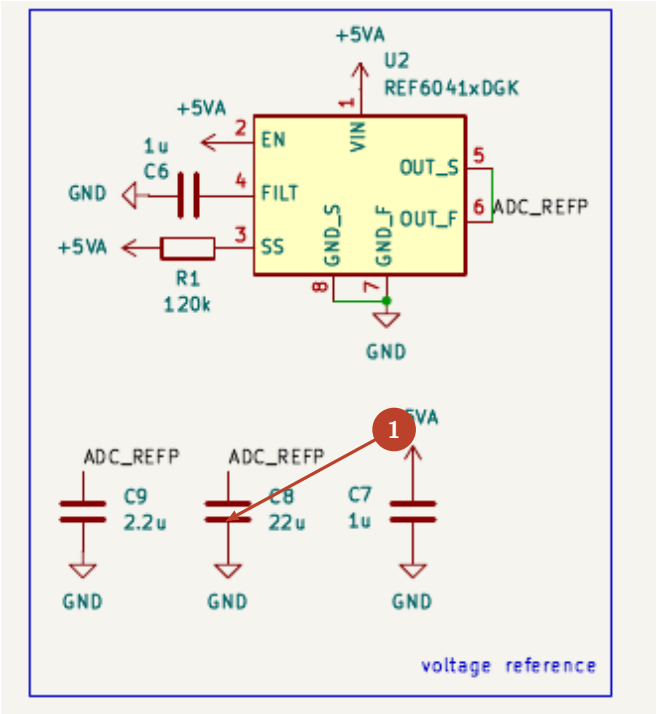
Connector contract to agree with Isaac.

- Four differential pairs, channel order and polarity; common-mode direction and return paths.
- I2C SCL/SDA, reset, voltage levels and pull-up ownership. Isaac's header spelling and Control-sheet connectivity also require correction.
- Power-rail names, directions, current limits and connector orientation. The analog regulator headroom issue must be fixed before relying on its 5 V output.
- This revision is receive-only. Transmit sequencing, ARM/interlock and transmit-watchdog implementation are outside this review's scope.

Acceptance. Compare a numbered pin table on both projects, then verify end-to-end net membership through a documented cable/connector mapping. Complete physical layout only after this agreement.

D5 / Reference and recording qualification

Checks requiring evidence, not asserted failures



1 — Verify reference stability. U2 drives C8 = 22 μF and C9 = 2.2 μF directly on ADC_REFP. REF60xx stability depends on effective output capacitance and ESR. Check actual parts, DC-bias derating, parallel capacitance and routing against TI's stable region; add controlled resistance if needed. No oscillation is claimed from the schematic alone. **Also check** SS current-limit configuration, worst-case reference load, settling and noise with the ADC running. R1 goes to +5VA; do not assume a resistor-to-ground example has the same behavior. SS pull-up is not classified here as a proven defect.

Power and BOM. Budget Daisy's 3.3 V output for microSD peaks, U3, translators and other loads. Select an exact 1.8 V-compatible 65.536 MHz oscillator, including jitter/startup ratings. U3's 1 μF input/output capacitors are nominally consistent with LP5907 guidance; verify effective capacitance and dissipation.

Recording proof required by the local spec. Four channels × 512,000 samples/s × 3 bytes = 6.144 MB/s packed payload, before framing and filesystem overhead. Qualify at least 12.288 MB/s sustained writes, a 16 MiB ring plus DMA buffers, a 0.5 s writer stall, and 60 minutes with no dropped/repeated samples or overruns. The SDMMC wiring alone does not demonstrate these targets.

Checks that looked reasonable. U4/U5 direction straps match register/control traffic; the separate data path is still missing. SDMMC DAT0–3/CMD/CLK map to Daisy physical pins 5/4/3/2/6/7 respectively. CAPA = 10 μF and CAPD = 2.2 μF match the ADC bypass table. These are schematic checks, not timing or hardware validation.

Sources: TI REF60xx, TI LP5907, TI SN74AVC4T245, and the local hydrophone-front-end discussion spec.